The limit of a function

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1) A new mathematical operation: limit of a function

We are familiar with summation, subtraction, division, multiplication, logarithm, trigonometric operations and many others since a young age, but although introduced to us later on, the limit of a function is perhaps the most important operation to set the foundation of calculus. Differently than the formerly mentioned, the limit take as input not a number but a function. To understand the concept of a limit, it is important to note one property inherent of all mathematical operations defined in the scope of our mathematics. Particularly within the scope of real numbers, the numerical patterns created by each and all individual operations that take numbers as input and give numbers as output have a very important property in common: **as the values of input used get closer and closer to a given arbitrary value, taken here as reference, the output value also gets indefinitely closer to the output of the reference input value**. No matter the sensibility of the operation, i.e., no matter the discrepancy between the Δouput of two values of input separated by a given Δinput, it is always possible to determine an interval of the domain of the operation where this **"proximity" property** holds as x approaches a given reference value chosen.[1] This can be regarded as something that "happens to be" a property intrinsic of all mathematic operations, as we defined, in the scope of real numbers. In other words, **it is a**

[1] It is not necessarily that this feature is not observed for other set of numbers, but if this set of number does not cherish that infinitude of numbers within an interval, the interval size between valid inputs may be too big for this property to be observed in the scope of the operations in hand. Each operation has an interval of a given length for which this can be observed, meaning that it is not for all of them that this behaviour holds on steadily.

property that is observed or verified to always occur, and not deduced from theory.[2]

Practical example 1: Imagine that you want to divide an input by 3. If you set an input equals to 6 as reference, you will get values closer and closer to 2 the more the input gets indefinitely closer to 6. Note that this is the case either for inputs that approach 6 either being smaller than it or larger than it. Note also that this conclusion holds no matter what operation or reference you consider.

It is not difficult to see that given that as this is a property inherent to the very definition of each and all operation, it is also necessarily a property automatically inherent to a whatever mathematical expression (i.e. whatever arrangement of operations). Thus, this is naturally carried up to functions. And it is in the context of this previously mentioned property of "proximity" for real functions, that the concept of limit arises:

Conceptually, the limit of a function is the "first" value of which the DV gets closer, without ever reaching, as the value of the IV increases or decreases towards a value, a (indefinite increase or decrease is also possible, as we will see later on). **It is therefore an information not about the value of the function at the given point but rather about the behaviour of the algebraic pattern in the surroundings of a point where $x = a$.**

For 1-IV functions, the limit operation is denoted algebraically by the syntax below, where the arrow functions as a symbol for "tends to":

$$\lim_{x \to a} f(x)$$

The most evident way, that perhaps comes first in mind, for actually determining this limit value is to actually calculate the value of the function for $x = a$, $f(a)$. **This happens as one realizes, given the "proximity" property of a real function previously mentioned, that conceptually the value of a limit matches what happens to be the value of the function for $x = a$.** After all, as discussed, it is always possible to find a value of $f(x)$ closer to $f(a)$ by using a value of x closer to a.

Despite apparently simple, the analysis of a limit is not always as straightforward to the point where calculating $f(a)$ always works. The two most obvious reasons for the non-straightforwardness within this discussion are: (i) the value of $f(x)$ for $x = a$ may be undefined; and (ii) it can be the case that the function is piecewise, and the value of a corresponds to the precise mark in which the change in governing expression happens, case in which the function may tend to different values depending if x is increasing *or* decreasing towards a (e.g. piecewise function). These are complications that encourage us to turn the brain on and

[2] Curiosity: another example of something that has is observed to always occur is the conservation of energy: no one ever proved that energy conserves. Yet, as no one ever observed that it doesn't, it became a law (rather than a theorem).

carry out the analysis mindfully rather than just automatically. Section 2 and 3 below shed light on what to do in these two scenarios as well as present another scenario in which the limit cannot be determined by computing $f(a)$.

Furthermore, a limit can also be evaluated as x grows or decreases indefinitely ($x \to +\infty$ or $x \to -\infty$), case in which evidently it is not even possible to evaluate the value of the function *per se*, since $\pm\infty$ is not a number. In this case, evaluation of the limit becomes something philosophical in nature: the result of operations featured in the expression of the function are replaced by values that correspond to what it should tend to without ever reaching as $x \to \pm\infty$. In other words, these values are "tendencies".

2) Side limits

Side limits have the same underlying definition previously mentioned of limits, but they consist on analysing the behaviour of the function only in one of the two neighbour sides of $x = a$: either (i) for values of the x that are approaching the target value, a, but are smaller than it, denoted as $x \to a^-$, or (ii) for values of x that are approach the target value, a, but are larger than it, denoted as $x \to a^+$. The notion of side limits comes in hand in scenarios where there is a reason or possibility for the function to have a different limit when $x \to a^-$ and when $x \to a^+$. In fact, in these scenarios, it is technically wrong to talk about a single limit value as defined in the previous section. **There are two more relevant types of scenarios in which side limit analysis may come in hand:** (1) when the expression of the function for $x = a$ leads to a non-existent value (e.g. the division of a number by 0); and (2) when the function is composed by more than one argument and the point of splitting between them is exactly the one where $x = a$.[3] The relevance of side limits for cases 2 is evident, while for cases 1 it may be regarded as something "seen to be necessary" as in principle the function can behave differently in the two neighbourhoods of $x = a$.

Finally, it may be worth mentioning that the protocol of merely looking for side limits is not really unique of these two previously mentioned cases: the concept itself can in principle be applied to any whatsoever limit calculation – they should lead to an answer upon reasoning within the specifics of the mathematical operations involved in the expression. In these cases, the splitting of the concept of limit into two should, when the two pieces of information are put together, give back that of a limit as initially introduced if the values are the same, given that the concepts are linked/overlap.

3) Special cases

In cases where $f(a)$ is undefined, an attempt to determine the limit of the function may be made by reliably analysing any trend in the expression of $f(x)$ as $x \to a$ and ultimately asserting a value for the limit based on this analysis (a practical example on how this method would work is given in section 4b). Yet this is not

[3] Note that some functions with apparently one mathematical expression can be split into two, such as $f(x) = |x|$.

always possible: the expression of the function may culminate on an indeterminate form. An indeterminate form within mathematics is, by definition, when the result of the limit of a function cannot be reliably determined because the mathematical operation(s) involved in the expression of the function ultimately either makes it impossible to safely assert about a value – in fact, indeterminate forms can be reached even in cases where $f(a)$ is defined.

a) The function yields an indeterminate form at $x = a$ or when $x \to \pm\infty$

There are seven cases of indeterminate forms. Specifically, five of them occur when the limit evaluation leads to indefinite growth or decay of parts of the function: they are $\pm\infty/\pm\infty$, $+\infty - \infty$, $0 * (\pm\infty)$, $\pm\infty^0, 1^{\pm\infty}$. The $\pm$ sign indicates that the infinite can indicate indefinite increase or decrease. These are indeterminate forms because the operations **and** the trends being operated (indefinite decrease or increase) ultimately causes conflicting behaviour: it is not possible to reliably infer what is the value $f(x)$ approaches without ever reaching. It is very easy to grasp why the first two of these five indeterminate forms indeed lead to an indeterminate limit. However, the fact that $1^{\pm\infty}$, for example, is an indeterminate form may be more difficult to grasp at first. In this matter, it might be helpful to note that it is indeed only an indeterminate form if the 1 is a "tendency" just like the $\pm\infty$, and not a constant: e.g., if we were to evaluate the limit of the function $f(x) = 1^x$ as $x \to \infty$, the answer is 1. By a similar reasoning, $0 * (\pm\infty)$ and $\pm\infty^0$ should only be an indeterminate form if the 0 is a trend as x goes to a given value, and not a constant.

Finally, there is also two other indeterminate forms: 0/0 and 0^0, completing the 7 indeterminate forms there are. The understanding on why 0^0 is indeterminate forms follow the same reasoning as mentioned in the previous paragraph: i.e. it should be noted that the 0 featured in them might not be literally a 0. Similarly, 0/0 is an indeterminate form because it is not possible to figure out by reasoning within the definition of the division operation if the result is a large number, a small number of a specific number. Interestingly, a function such as x^x is actually defined at $x = 0$ in the scope of the definition of exponentiation: at $x = 0$, $x^x = 1$. Unlike 0^0, a function that yields 0/0 for $x = a$ is indeed undefined there, with the reason being indetermination, as defined in a previous document.

As seen above, indeterminate forms can occur in cases where the function is undefined as well as in cases where the function is defined. Notably, not having the function be defined at $x = a$ doesn't mean that the limit isn't defined either. **As previously mentioned, the limit, on its definition, is an information concerning the behaviour of the values of $f(x)$ on the neighbourhood of $x = a$, being it a question fundamentally different than calculating the value of $f(a)$ itself**: calculating the function at $x = a$ came up simply as one reliable methodology to do so. Similarly, reaching a mathematical expression that does not allow to safely state the limit of $f(x)$ as x displays a given trend doesn't necessarily mean there isn't a limit.

As it turns out, determining the limit of a function $f(x)$ when an indeterminate form is reached is a scenario in which the concept of "semi-equivalent" and equivalent functions come in hand. If $f(x)$ and an equivalent function share all the solutions, it is certain that they share the same limit as x tends indefinitely to a given value: after all, the values of the function are the same for any x. This allows to use the equivalent function to determine the limit of $f(x)$ as we fall in the case (i) mentioned in the previous paragraph. **Similarly, if $f(x)$ and a semi-equivalent function, defined here as $g(x)$, share the solutions except for $x = a$, it is certain that they share the same tendency as $x \to a$ anyways: after all, all the other values of the functions are precisely the same for any other x.** So, if unlike $f(x)$, $g(x)$ is defined at $x = a$, the methodology of calculating its value at $x = a$ and claiming it to be the limit of $f(x)$ as $x \to a$ can be used. This then validates the notion of seeking a "semi-equivalent" function to determine the limit of a given function in the scenario where an indetermination exists (case (ii) mentioned in the previous paragraph). As the methodology is validated, it becomes a matter of being intelligent enough to find the equivalent or semi-equivalent function within the scope of the mathematical operations involved in the expression of $f(x)$.

b) A value for the function doesn't exist at $x = a$ but no indeterminate form is obtained

When the value of a function doesn't exist for a given input, it ties back to the operations composing the expression. The few cases in which an output doesn't exist for a given input within a given mathematical operation (e.g. division of a number by 0, logarithm of 0, tangent of 90°…) come from the definition of the algorithm of the given operation. From knowing the algorithm of the most common mathematical operations, it is possible to conclude that often (but not necessarily always) these functions have a diverging behaviour near the value of x for which the output doesn't exist. This causes the function (just like the result of the mathematical operation causing this scenario *per se*) to decrease or increase indefinitely as $x \to a$. In other words, instead of an indeterminate form, a limit for the function equal to $\pm\infty$ is often reached when $f(a)$ doesn't exist. This means it does not make sense to look for an equivalent or semi-equivalent function: the value of the limit can be successfully and **clearly** determined from expression of the function. Here is a practical example on how this scenario differs from those of indeterminate forms:

Practical example: Consider the functions $f(x) = (x^2 - 4)/(x - 2)$ and $g(x) = 2/(x - 2)$. When $x \to 2$ in the case of $f(x)$, not only we observe an indeterminate form when replacing $x = 2$, but we also are not able to deduce anything certain about a trend when analysing what happens to $f(x)$ when $x \to 2$: **since both the numerator and the denominator decrease indefinitely, the algorithm of the division operation does not allow to safely state a trend. Since the expression of $f(x)$ does not allow any conclusions via these two point of analysis, we then search for a semi-equivalent function.** On the other hand, in the case of $g(x)$, it is not possible to find a value for $g(2)$ as a division by 0 is non existent but the expression of the function allows to reliably evaluate the trend of the function as $x \to 2$: in the scope of the algorithm of division, the value of $g(x)$ has to necessarily become indefinitely large or indefinitely small depending on whether x approaches 2 from larger or smaller values (side limit analysis is needed). **As a clear conclusion can be reached in a solid way, there is no need to search for another function as in the former case.**

4) Properties of limits as mathematical operations

This section aims to **state** some properties of functions, $f(x)$, that can be written as a function of two other functions, $g(x)$ and $h(x)$. A proof can be found in textbooks or elsewhere. We provide here only the statements and a reliable way to see their validity that is not rooted on the usual equation-based demonstration. Given $f(x)$, $g(x)$ and $h(x)$, we may be able to write (exceptions are discussed below):

- For $f(x) = g(x) + h(x)$, $\lim_{x\to a} f(x) = \lim_{x\to a}\big(g(x) + h(x)\big) = \lim_{x\to a} g(x) + \lim_{x\to a} h(x)$
- For $f(x) = g(x) - h(x)$, $\lim_{x\to a} f(x) = \lim_{x\to a}\big(g(x) - h(x)\big) = \lim_{x\to a} g(x) - \lim_{x\to a} h(x)$
- For $f(x) = g(x).h(x)$, $\lim_{x\to a} f(x) = \lim_{x\to a}\big(g(x).h(x)\big) = \lim_{x\to a} g(x)\,.\lim_{x\to a} h(x)$
- For $f(x) = g(x)/h(x)$, $\lim_{x\to a} f(x) = \lim_{x\to a}\big(g(x)/h(x)\big) = \lim_{x\to a} g(x)\,/\lim_{x\to a} h(x)$ provided that $\lim_{x\to a} h(x) \neq 0$.

The validity of this property can be reliably verified by realizing that the very foundation that allows to define the limit of a function as a mathematical operation is the fact that summation, subtraction, multiplication and division share the "proximity" property previously mentioned. If the value of $f(x)$ for any x within the interval immediate to $x = a$ can be written as a function of the values of $g(x)$ and $h(x)$, the limit of these two latter functions as $x \to a$ correspondingly operated in the way $f(x)$ depends on $g(x)$ and $h(x)$ should correspond to the limit of $f(x)$ as $x \to a$. Clear exceptions to the previous reasoning are if the limit of both $g(x)$ and $h(x)$ are defined but lead to an indeterminate form at $x = a$, case in which determining the limit of $f(x)$ requires finding an alternative function as discussed above.

5) Limits of functions of more than 1 IV

In functions of more than one IV, the value of the DV depends on the value of more than one IV. In their context, the proximity property, the definition of limit, its particularities and the ways to compute it remain the same as discussed above. The particularities in the case of more functions of more than 1-IV are firstly that the limit can be calculated for more than one IV. In this case, the tendencies are evaluated separately and subsequently for each given IV within a given order. Secondly, the result of a limit with respect to one IV in the case of multi-IV functions is not a numerical result but another function as a response. This resulting function should correspond to the value of the limit for all possible values of the IVs that remain in the expression as the given IV tends to the given target in the framework of the original function.

Practical example: Given the function $z = f(x, y) = x^2 + y$, we have that the limit of $f(x, y)$ as $y \to 0$ is always going to be given by $f(x) = x^2$ for all x.

Putting together the first and second particularities previously mentioned, one can conclude that when the limit of several IVs are evaluated subsequently, the resulting function (or value, if the limit with respect to all IVs are calculated) is the same regardless of the order considered. This is necessary for mathematical coherence: the values of the function remain the same regardless of the order in which specific IV values are replaced in the argument of $f(IVs)$, and so must be their tendencies.